BIOSTATISTICS AND EPIDEMIOLOGY

Process of Measuring & Analysing Health Data

TOPIC OVERVIEW

LEARNING OBJECTIVES

- Distinguish a proportion from a rate, and explain why time changes everything.
- Compute all 5 measures of morbidity from a single outbreak-and-cohort scenario.
- Select the correct numerator and denominator for 11 mortality indicators.
- Calculate 5 natality measures and connect them to a population's growth

Every case scenario in this lecture pulls from the table below: the Municipality of Santa Cruz, calendar year 2025. Your job each time is to pick the right rows; not every row is used by every formula.

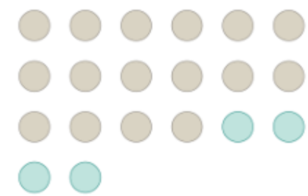
Mid-year population (total)	50,000	Neonatal deaths (<28 d)	9
Population aged 65+	3,500	Postneonatal deaths	6
Women aged 15–49	12,500	Fiesta attendees (outbreak)	300
Women aged 20–24	2,100	New GI cases in 24 h	45
Live births	750	Household contacts exposed	150
Live births < 2,500 g	54	Secondary GI cases	27
Births to women 20–24	210	Dengue cases reported	500
Fetal deaths (≥20 wks)	12	Deaths among dengue cases	15
Total deaths, all causes	380	Diabetic, point survey (Jul 1)	620
Deaths, age 65+	210	Diabetic, any time in year	900
Deaths, cardiovascular	95	Cohort enrolled (HTN study)	500
Deaths, cancer	70	Person-years accumulated	2,150
Maternal deaths	2	New HTN cases in cohort	86

I. FREQUENCY MEASURES

- The building blocks: proportion and rate.
- Get this distinction right, and every measure that follows is just a variation on it.

A. PROPORTION

- A **part expressed against its whole** — the numerator is a subset of the denominator.
- No time dimension. It is a snapshot ratio, not a speed.
- Usually expressed as a percentage.
- $\frac{\mathit{Number\ with\ the\ characteristic}}{\mathit{Total\ number\ in\ the\ group}} \times 100$

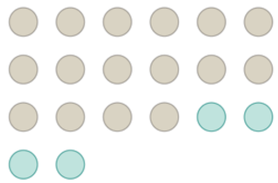


Case Scenario

- Of Santa Cruz's 50,000 mid-year residents, 3,500 are aged 65 or older. What proportion of the population is elderly?
  - Population aged 65+ = 3,500
  - Total mid-year population = 50,000
  - 3,500 ÷ 50,000 × 100 = **7% of the population**

B. RATIO

- The relative magnitude of two quantities or a **comparison of any two values**
  - Not talking about the whole
- $\frac{\mathit{Number\ of\ persons\ etc.\ in\ one\ group}}{\mathit{Number\ of\ persons\ etc.\ in\ another\ group}}$



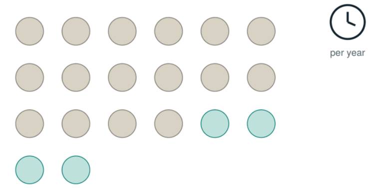
- For example, in a block, there are 30 females and 10 males, therefore the ratio of male to females is 10:30
  - Read as *for every 10 males there are 30 females*
  - Can be expressed in lower numbers or simplified into *for every one (1) male, there are three (3) females*
- The first number asked, should be the first number presented (NOT INTERCHANGEABLE)

Case Scenario

- During 2025, Santa Cruz recorded 95 deaths due to cardiovascular causes and 70 deaths due to cancer. Express this as a ratio.
  - Deaths, cardiovascular = 95
  - Deaths, cancer = 70
  - 95 ÷ 70 = **1.36**

C. RATE

- Measures **how fast an event occurs in a population** — always carries a time unit.
- Numerator = events during a period; denominator = population at risk during that same period.
- This is the feature every rate in this lecture will share.
- $\frac{\textit{Number of events in a period}}{\textit{Population at risk during that period}} \times 10^n$



- 10^n is a variable or multiplier that can be adjusted depending on how rare the occurrence is for better visualization.
  - For very rare cases, the result of the formula may produce an extremely small decimal (e.g., **0.00006**).
  - If we express this as **0.00006 cases per 1,000 population per year**, it can be difficult to visualize or interpret.
  - Instead, we can adjust the multiplier to obtain a whole number. For example: **0.00006 × 10,000 = 6**
  - Therefore, this can be expressed as **6 cases per 10,000 population per year**, which is easier to appreciate and interpret
  - Rarer case = smaller decimal = bigger multiplier

Case Scenario

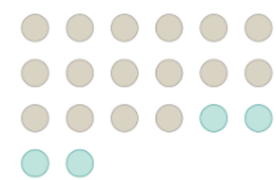
- During 2025, Santa Cruz recorded 500 new dengue cases among its 50,000 residents. Express this as a rate.
  - New dengue cases in 2025 = 500
  - Mid-year population = 50,000
  - 500 ÷ 50,000 × 1,000 = **10 cases per 1,000 population per year**

II. MEASURES OF MORBIDITY

- New illness, existing illness, snapshots, and follow-up — five ways to count who's sick and when.
  - No Death

A. INCIDENCE PROPORTION (ATTACK RATE)

- The **proportion of an at-risk group that develops new disease over a defined, usually short, period.**
- Common in outbreak investigations — hence "attack rate."
- Despite the name, it is technically a proportion: no person-time in the denominator
- Commonly used during outbreaks (not permanent but cases will rise randomly, seasonal)
- $\frac{\textit{New cases during the period}}{\textit{Population at risk at the start}} \times 100$



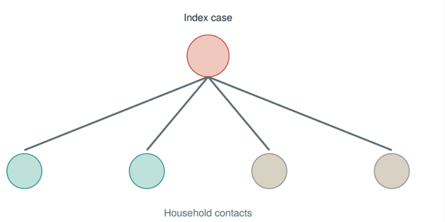
- *"Among those who were at risk, who actually got sick?"*
- Considered as your "Primary Attack Rate"

Case Scenario

- At the Barangay Mabini fiesta on June 14, 300 people attended. Within 24 hours, 45 developed gastroenteritis.
  - Fiesta attendees = 300
  - New GI cases within 24 h = 45
  - 45 ÷ 300 × 100 = **15% attack rate**

B. SECONDARY ATTACK RATE

- Measures **person-to-person spread within a close-contact group,** like a household or classroom.
- The denominator excludes the primary (index) case(s) — only susceptible contacts count.
- A high SAR signals efficient transmission once exposure has already happened once.
- $\frac{New\ cases\ among\ contacts}{Contacts - primary\ case(s)} \times 100$



- *"How efficient is the transmission?"*
- **Contact tracing** may be done

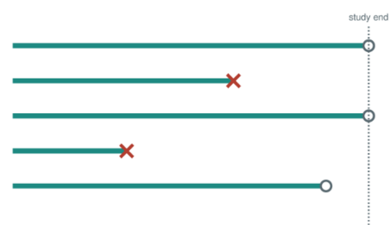
Case Scenario

- The 45 fiesta cases went home to their households. Investigators traced 150 household contacts who had not attended. Of those, 27 developed gastroenteritis.
  - Household contacts traced = 150
  - Secondary cases among contacts = 27
  - 27 ÷ 150 × 100 = **18% secondary attack rate**

Difference Between Primary and Secondary Attack Rate	
Primary Attack Rate	**Secondary Attack Rate**
How many are infected from the entire population at risk?	From one case(primary), how efficient is the disease being transmitted to other people (secondary).

C. INCIDENCE RATE (PERSON-TIME RATE)

- Denominator is **person-time at risk**, not head count — it accounts for people entering, leaving, or being followed for different lengths of time.
  - Time is involved
- Standard for cohort studies with long or uneven follow-up.
  - Used for long-term studies to count the years when the study started until the person either died, got healed, how many years that person chose to join, and when the study ends.
- A true rate: has both an event count and an explicit time unit.
- $\frac{New\ cases\ during\ follow-up}{\Sigma\ person-time\ at\ risk} \times 1000$

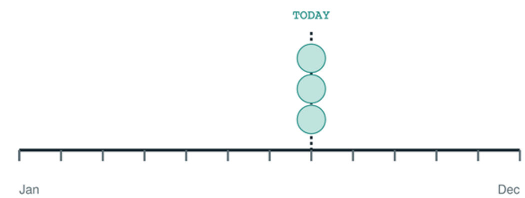


Case Scenario

- A 5-year hypertension cohort enrolled 500 Santa Cruz residents. Because people joined, dropped out, and were diagnosed at different points, the study accumulated 2,150 person-years of observation. 86 new hypertension cases were diagnosed.
  - New HTN cases = 86
  - Person-years accumulated = 2,150
  - 86 ÷ 2,150 × 1,000 = **40 cases per 1,000 person-years**

D. POINT PREVALENCE

- The proportion of a population with a disease **at one specific moment in time.**
- Counts everyone currently affected — old cases and new cases alike — as of that single day.
- A snapshot, not a rate: no follow-up time in the denominator.
- $\frac{Existing\ cases\ at\ that\ moment}{Population\ at\ that\ same\ moment} \times 1000$



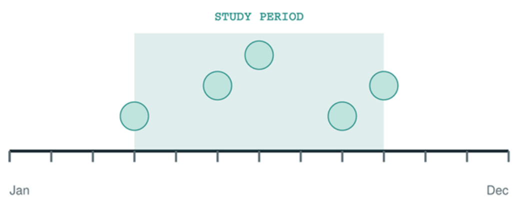
Case Scenario

- A diabetes registry check on July 1, 2025 found 620 Santa Cruz residents currently living with diabetes, regardless of when they were first diagnosed.
  - Diabetic as of July 1 = 620
  - Population on July 1 ≈ 50,000
  - 620 ÷ 50,000 × 1,000 = **12.4 per 1,000 population**

E. PERIOD PREVALENCE

- The proportion of a population that **had the disease at any time during a defined window.**

- Wider than point prevalence — anyone affected at any point in the stretch counts, not just on one day.
- Denominator typically uses the average or mid-year population.
- $\frac{\textit{Cases existing during the period}}{\textit{Average population}} \times 1000$



Case Scenario

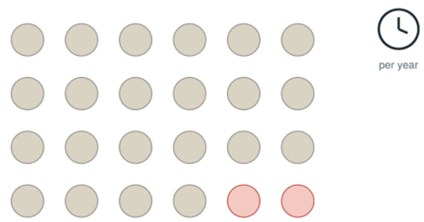
- Reviewing the full year, records show 900 Santa Cruz residents were living with diabetes at some point between January and December 2025.
    - Diabetic at any point in 2025 = 900
    - Average population ≈ 50,000
    - 900 ÷ 50,000 × 1,000 = **18 per 1,000 population**

III. MORTALITY INDICATORS

- Eleven measures, one recurring question: deaths from what, among whom, out of what denominator?

A. MORTALITY RATE/CRUDE DEATH RATE

- **Every death, from every cause, counted against the entire population.**
- "Crude" means unadjusted — it ignores age structure entirely.
- The baseline every other mortality measure narrows down from
- Time element is involved
- $\frac{\textit{Total deaths}}{\textit{Mid-year population}} \times 1000$

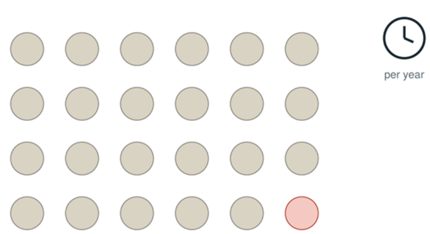


Case Scenario

- Santa Cruz recorded 380 deaths from all causes in 2025, against a mid-year population of 50,000.
    - Total deaths, all causes = 380
    - Mid-year population = 50,000
    - 380 ÷ 50,000 × 1,000 = **7.6 deaths per 1,000 population**

B. CAUSE-SPECIFIC DEATH RATE

- Same denominator as CDR — the whole population — but the numerator narrows to one named cause.
- Lets you **compare the burden of specific diseases across populations or years.**
- Because the numerator is small, it's usually scaled per 100,000.
- $\frac{\textit{Deaths from one named cause}}{\textit{Mid-year population}} \times 100000$

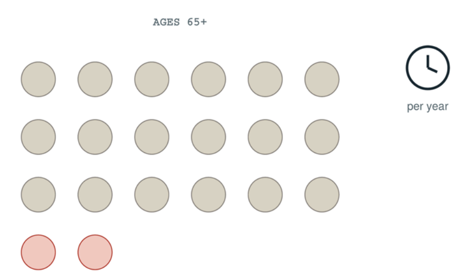


Case Scenario

- Of the 380 total deaths in Santa Cruz, 95 were attributed to cardiovascular disease.
    - Cardiovascular deaths = 95
    - Mid-year population = 50,000
    - 95 ÷ 50,000 × 100,000 = **190 per 100,000 population**

C. AGE-SPECIFIC DEATH RATE (ASDR)

- Restricts both the numerator and the denominator to a **single age band.**
    - Pinpoints which age-group contributes more to the death rate
- Different from cause-specific DR: there the population stayed whole and the cause narrowed; here the cause stays open and the population narrows.
- Corrects for the fact that crude rates can be misleading across differently-aged populations.
- $\frac{\textit{Deaths within one age group}}{\textit{Population of that same age group}} \times 1000$



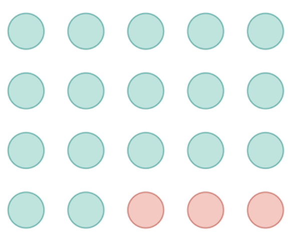
Case Scenario

- Among Santa Cruz's 3,500 residents aged 65 and above, 210 deaths (from any cause) occurred in 2025.
    - Deaths, age 65+ = 210
    - Population aged 65+ = 3,500

- 210 ÷ 3,500 × 1,000 = **60 deaths per 1,000 population aged 65+**

D. CASE FATALITY RATE (CFR)

- The denominator is no longer the general population — **only people who already have the disease.**
- Answers: "*If you catch this disease, what are your odds of dying from it?*"
    - Higher CFR means there's a higher chance of dying from the disease
    - Helps in predicting patient's lifetime
- A measure of a disease's severity, not its spread.
- Counts the entire recorded cases with no timeframe included
- $\frac{\textit{Deaths from the disease}}{\textit{Diagnosed cases of the disease}} \times 100$

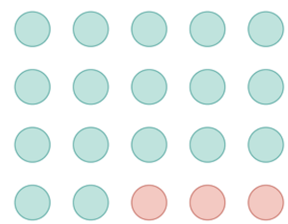


Case Scenario

- Santa Cruz's surveillance unit reported 500 dengue cases in 2025. Of those diagnosed cases, 15 died from dengue.
    - Dengue cases diagnosed = 500
    - Deaths among dengue cases = 15
    - 15 ÷ 500 × 100 = **3% case fatality rate**

E. DEATH-TO-CASE RATIO

- Ties deaths from a disease to cases reported in the **same period** — not necessarily one cohort followed from diagnosis to outcome.
- The arithmetic often looks identical to CFR; the justification for pairing the numbers is what differs.
- Frequently used in routine surveillance reporting, where cases and deaths are simply tallied for the same window.
- $\frac{\textit{Deaths from disease (period)}}{\textit{New cases reported (same period)}} \times 100$

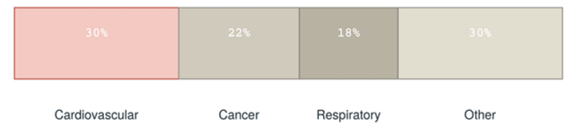


Case Scenario

- Using the same dengue surveillance log: 500 cases were reported in 2025, and 15 deaths among dengue patients were logged in that same year.
    - Dengue cases reported in 2025 = 500
    - Dengue deaths reported in 2025 = 15
    - 15 ÷ 500 × 100 = **3% — same figure as CFR, different justification**

F. PROPORTIONATE MORTALITY

- Slices up the **deaths that already happened, by cause** — has no connection to the size of the living population at all.
    - "*Among all deaths, how many died of that specific cause?*"
- Tells you what share of all deaths a cause represents, not the risk of dying from it.
- Do not confuse with cause-specific death rate — the denominators are completely different.
- $\frac{\textit{Deaths from one cause}}{\textit{Deaths from all causes}} \times 100$



Case Scenario

- Of the 380 total deaths recorded in Santa Cruz in 2025, 95 were due to cardiovascular disease.
    - Cardiovascular deaths = 95
    - Total deaths, all causes = 380
    - 95 ÷ 380 × 100 = **25% of all deaths were cardiovascular**

G. NEONATAL MORTALITY RATE

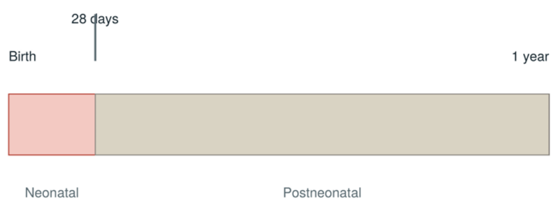
- Deaths in the **first 28 days of life only** — the earliest and most fragile window of infancy.
- The denominator is live births, not population — it's specific to the birth cohort.
- High neonatal mortality often points to complications around delivery itself
- $\frac{\textit{Deaths of babies under 28 days old}}{\textit{Live births}} \times 1000$

Case Scenario

- Of Santa Cruz's 750 live births in 2025, 9 infants died before reaching 28 days of age.
  - Deaths under 28 days = 9
  - Live births = 750
  - 9 ÷ 750 × 1,000 = **12 per 1,000 live births**

H. POSTNEONATAL MORTALITY RATE

- Covers the remainder of the first year: **28 days up to (but not including) 1 year of age.**
- Same denominator as neonatal mortality rate — live births — just a different slice of the first year.
- Deaths in this window more often reflect the home and community environment than delivery complications
- $\frac{Deaths, 28\ days\ to\ 1\ year}{Live\ births} \times 1000$

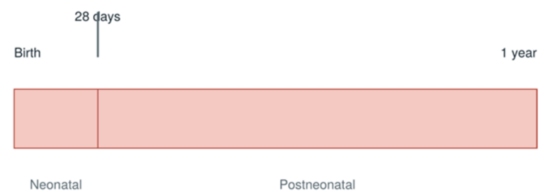


Case Scenario

- Of the same 750 live births, 6 additional infants died sometime between 28 days and their first birthday.
  - Deaths, 28 days–1 year = 6
  - Live births = 750
  - 6 ÷ 750 × 1,000 = **8 per 1,000 live births**

I. INFANT MORTALITY RATE

- Covers the **entire first year of life**: *neonatal + postneonatal deaths combined.*
- One of the most widely used indicators of a population's overall health and socioeconomic status.
- Same denominator throughout this family of measures: live births
- $\frac{Deaths\ under\ 1\ year\ old}{Live\ births} \times 1000$

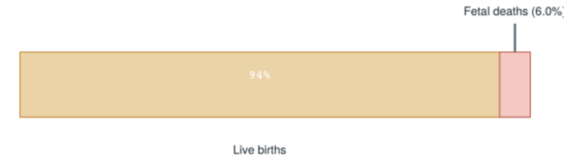


Case Scenario

- Combining the two previous slides: 9 neonatal deaths plus 6 postneonatal deaths, out of 750 live births.
  - Neonatal + postneonatal deaths = 9 + 6 = 15
  - Live births = 750
  - 15 ÷ 750 × 1,000 = **20 per 1,000 live births**

J. FETAL DEATH RATE

- Covers fetal deaths at **20 weeks gestation or later.**
  - Has not yet been given birth to
  - Counts any cause as long as the fetus dies
- The one exception in this family: the denominator is NOT live births alone — it's **live births plus fetal deaths.**
- $\frac{Fetal\ deaths\ (\geq 20\ weeks)}{Live\ births + fetal\ deaths} \times 1000$

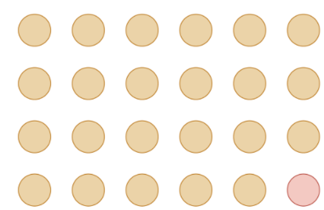


Case Scenario

- Alongside the 750 live births, Santa Cruz recorded 12 fetal deaths at or beyond 20 weeks gestation in 2025.
  - Fetal deaths (≥20 wks) = 12
  - Live births + fetal deaths = 750 + 12 = 762
  - 12 ÷ 762 × 1,000 ≈ **15.7 per 1,000 total births**

K. MATERNAL MORTALITY RATE

- Deaths from **pregnancy-related (puerperal) causes,** against live births as a proxy for "women at risk."
  - The mother dying of other causes does not count
- A genuinely rare event at the population level — that rarity is exactly why the multiplier is so large.
- Scaling by 100,000 instead of 100 or 1,000 turns a tiny fraction into a readable whole number.
- $\frac{Pregnancy{-}related\ deaths}{Live\ births} \times 100000$



Case Scenario

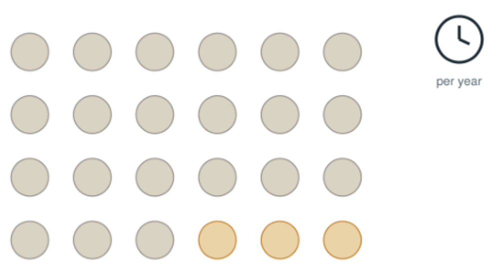
- Santa Cruz recorded 2 maternal deaths in 2025, against 750 live births.
    - Maternal deaths = 2
    - Live births = 750
    - 2 ÷ 750 × 100,000 ≈ **266.7 per 100,000 live births**

IV. NATALITY (BIRTH) MEASURES

- Five measures that describe how a population grows from the crudest count of births to a single age band of mothers.

A. CRUDE BIRTH RATE

- Every live birth counted against the entire population, regardless of age or sex.
- "Crude" for the same reason as CDR: no adjustment for who is actually capable of giving birth.
    - How many babies were born in a population
- The natality equivalent of crude death rate — same logic, opposite direction.
- If this number is high that means that that specific population is actively growing in number (ppl r doing it alot)
- $\frac{Live\ births}{Mid-year\ population} \times 1000$

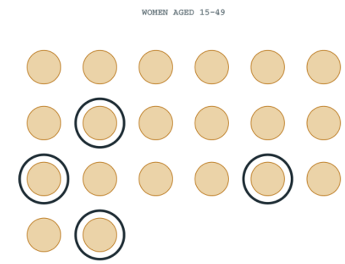


Case Scenario

- Santa Cruz recorded 750 live births in 2025, against a mid-year population of 50,000.
    - Live births = 750
    - Mid-year population = 50,000
    - 750 ÷ 50,000 × 1,000 = **15 births per 1,000 population**

B. GENERAL/CRUDE FERTILITY RATE

- Sharpens the denominator to **only women of childbearing age** (commonly 15–49).
    - As long as women are of child bearing age (15-49 y.o.) they are considered as part of the population regardless of how they identify themselves as (nun, gay, trans, etc.)
- A fairer measure than CBR, since men and children can't give birth but still sit in CBR's denominator.
    - Since women are the ones contributing to the population increase we use women of childbearing age instead of the whole population since the whole population includes men, women not of childbearing age etc.
- Same numerator as CBR — only the denominator changes.
- $\frac{Live\ births}{Women\ aged\ 15–49} \times 1000$

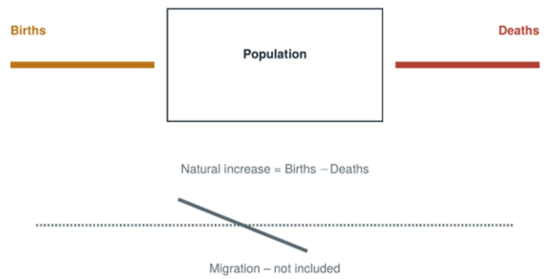


Case Scenario

- Those same 750 live births, now measured only against Santa Cruz's 12,500 women of childbearing age.
    - Live births = 750
    - Women aged 15–49 = 12,500
    - 750 ÷ 12,500 × 1,000 = **60 births per 1,000 women aged 15–49**

C. CRUDE RATE OF NATURAL INCREASE

- Simply CBR minus CDR — births in, deaths out.
    - How many are born and how many died
- Migration is deliberately excluded; this measures natural change only.
- A quick gut-check on whether a population is growing or shrinking from births and deaths alone
    - Natural *increase* indicates an increase in population.
- $CBR - CDR = per\ 1000$

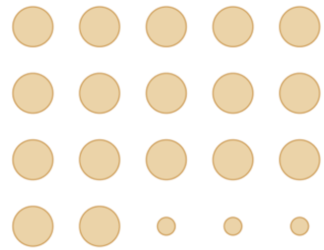


Case Scenario

- Santa Cruz's crude birth rate was 15 per 1,000 (Slide 4.1); its crude death rate was 7.6 per 1,000 (Slide 3.1).
  - CBR = 15 per 1,000
  - CDR = 7.6 per 1,000
  - 15 - 7.6 = **7.4 per 1,000 natural increase**

D. LOW-BIRTH-WEIGHT RATIO

- The share of live births weighing under 2,500 grams at delivery.
  - Usually the babies who are born premature (preemies)
  - Possibly caused by poor maternal nutrition, poor prenatal care, and born pre-term
- A widely used proxy for maternal nutrition, prenatal care access, and newborn risk.
- **Technically a proportion** — both numerator and denominator are drawn from the same year's live births.
  - Not a comparison of subsets, therefore, not a ratio
- Keyword: Babies who are born very light
- $\frac{\textit{Live births} < 2{,}500\ g}{\textit{Total live births}} \times 100$

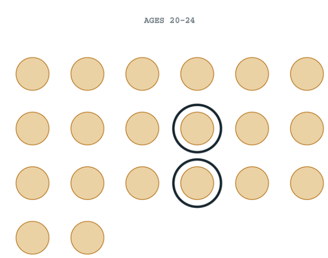


Case Scenario

- Of Santa Cruz's 750 live births in 2025, 54 weighed less than 2,500 grams at birth.
  - Live births < 2,500 g = 54
  - Total live births = 750
  - 54 ÷ 750 × 100 = **7.2% low birth weight**

E. AGE-SPECIFIC FERTILITY RATE

- Same logic as general fertility rate, narrowed to a single 5-year maternal age band.
  - There's an age bracket, e.g. 20-25, 15-19, 15-29 etc
- Lets you see which age groups are actually driving a population's birth rate.
- The natality mirror of age-specific death rate: both numerator and denominator restricted to one age group.
- $\frac{\textit{Births to women in one age group}}{\textit{Women in that same age group}} \times 1000$



SUMMARY TABLE

FREQUENCY			
Measure	**Numerator**	**Denominator**	**x**
Proportion	Part	Whole	100
Ratio	One group	Another group	10^n
Rate	Events	Population x Time	100

MORBIDITY			
Measure	**Numerator**	**Denominator**	**x**
Incidence Proportion (attack rate)	New Cases	Population at risk (start)	100
Secondary Attack Rate	New Cases among contacts	Contacts - primary Case	100
Incidence Rate (person-time)	New cases	Σ person-time at risk	1000
Point prevalence	Existing cases at a moment	Population at that moment	1000
Period prevalence	Existing cases in a period	Average population	1000

MORTALITY			
Measure	**Numerator**	**Denominator**	**x**
Mortality rate / CDR	Total deaths	Mid-year population	1000
Cause-specific death rate	Deaths from one cause	Mid-year population	100,000
Age-specific death rate	Deaths in one age group	Population of that age group	1000
Case fatality rate	Deaths from the disease	Diagnosed cases	100
Proportionate mortality	Deaths from one cause	Deaths from all causes	100
Death-to-case ratio	Deaths (period)	New cases reported (period)	100
Neonatal mortality rate	Deaths <28 days	Live births	1000
Postneonatal mortality rate	Deaths, 28 days - 1 year	Live births	1000
Infant mortality rate	Deaths < 1 year	Live births	1000
Fetal death rate	Fetal deaths (≥ 20 weeks)	Live births + fetal deaths	1000
Maternal mortality rate	Pregnancy-related deaths	Live births	100,000

MORBIDITY			
Measure	**Numerator**	**Denominator**	**x**
Crude birth rate	Live births	Mid-year population	1,000
General fertility rate	Live births	Women aged 15–49	1,000
Rate of natural increase	CBR - CDR	—	1,000
Low-birth-weight ratio	Live births < 2,500 g	Total live births	100
Age-specific fertility rate	Births in one age group	Women in that age group	1,000

WORKSHEET ANSWERS

Scenario 1

A wedding in Barangay Lupang Pangako serves shared food to 150 guests. Within 8 hours, 60 guests develop acute gastroenteritis.

- **Correct measure:**
  - *INCIDENCE PROPORTION*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{New\ cases\ during\ the\ period}{Population\ at\ risk\ at\ the\ start} \times 100$
- **Computation & answer:**
  - $\frac{60\ guests}{150\ guests} \times 100 = 40\%\ attack\ rate$

Scenario 2

A province with 3,200 live births in a year records 3 deaths among mothers from pregnancy-related complications.

- **Correct measure:**
  - *MATERNAL MORTALITY RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Pregnancy\ related\ deaths}{Live\ births} \times 100,000$
- **Computation & answer:**
  - $\frac{3\ maternal\ deaths}{3,200\ live\ births} \times 100,000$
  - $\approx 93.8\ per\ 100,000\ live\ births$

Scenario 3

A diagnostic laboratory tests 500 specimens for HIV antibody; 45 come back reactive. What share of specimens tested were reactive?

- **Correct measure:**
  - *PROPORTION*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Number\ with\ the\ characteristic}{Total\ number\ in\ the\ group} \times 100$
- **Computation & answer:**
  - $\frac{45\ reactive\ specimens}{500\ total\ specimens} \times 100 = 9\%\ of\ the\ specimens$

Scenario 4

Health center records show 310 residents were treated for tuberculosis at some point during the year, out of a catchment population of 60,000.

- **Correct measure:**
  - *PERIOD PREVALENCE*

- **Formula (numerator / denominator × multiplier):**
    - $\frac{\textit{Cases existing during the period}}{\textit{Average population}} \times 1,000$
- **Computation & answer:**
    - $\frac{310 \textit{ residents treated for tuberculosis during the year}}{60,000 \textit{ catchment population}} \times 1,000$
    - $\approx 5.2 \textit{ per } 1,000 \textit{ population}$

Scenario 5

A meningitis outbreak sickens 80 people in a district. Of those diagnosed, 6 die from meningitis.

- **Correct measure:**
    - *CASE FATALITY RATE*
- **Formula (numerator / denominator × multiplier):**
    - $\frac{\textit{Deaths from the disease}}{\textit{Diagnosed cases of the disease}} \times 100$
- **Computation & answer:**
    - $\frac{6 \textit{ deaths from meningitis}}{80 \textit{ cases of meningitis}} \times 100$
    - $= 7.5\% \textit{ case fatality rate}$

Scenario 6

A region's crude birth rate is 18 per 1,000 population and its crude death rate is 6 per 1,000 population for the same year.

- **Correct measure:**
    - *CRUDE RATE OF NATURAL INCREASE*
- **Formula (numerator / denominator × multiplier):**
    - $(\textit{Crude Birth Rate} - \textit{Crude Death Rate}) \textit{ per } 1,000$
- **Computation & answer:**
    - $18 - 6 = 12 \textit{ per } 1,000 \textit{ natural increase}$

Scenario 7

A daycare center has one confirmed measles case. Twenty-two other children who had close contact with that child were monitored; 8 of them went on to develop measles.

- **Correct measure:**
    - *SECONDARY ATTACK RATE*
- **Formula (numerator / denominator × multiplier):**
    - $\frac{\textit{New cases among contacts}}{\textit{Contacts} - \textit{primary case(s)}} \times 100$
    - *Note: include only the close contacts in the denominator.*
- **Computation & answer:**
    - $\frac{8 \textit{ new measles cases}}{22 \textit{ close contacts}} \times 100$
    - $= 36.3\% \textit{ secondary attack rate}$

Scenario 8

A tertiary hospital records 900 total deaths in a year. Of those, 180 were attributed to cancer.

- **Correct measure:**
    - *PROPORTIONATE MORTALITY*
- **Formula (numerator / denominator × multiplier):**
    - $\frac{\textit{Deaths from one cause}}{\textit{Deaths from all causes}} \times 100$
    - *Note: include only the close contacts in the denominator.*
- **Computation & answer:**
    - $\frac{180 \textit{ deaths due to cancer}}{900 \textit{ total deaths}} \times 100$
    - $= 20\% \textit{ of all deaths were due to cancer}$

Scenario 9

A lying-in clinic recorded 1,800 live births and 14 fetal deaths at 20 weeks gestation or later during the same year.

- **Correct measure:**
    - *FETAL DEATH RATE*
- **Formula (numerator / denominator × multiplier):**
    - $\frac{\textit{Fetal deaths } (\geq 20 \textit{ weeks})}{\textit{Live births} + \textit{fetal deaths}} \times 1,000$
- **Computation & answer:**
    - $\frac{14 \textit{ Fetal deaths}}{1{,}800 \textit{ Live births} + 14 \textit{ fetal deaths}} \times 1,000$
    - $\approx 7.7 \textit{ per } 1,000 \textit{ total births}$

Scenario 10

A hospital laboratory processes 12,000 tests in one month. Of these, 3 results were critical (panic) values requiring immediate physician notification.

- **Correct measure:**
    - *RATE*
- **Formula (numerator / denominator × multiplier):**
    - $\frac{\textit{Number of events in a period}}{\textit{Population at risk during that period}} \times 10^n$
- **Computation & answer:**
    - $\frac{3 \textit{ critical test values}}{12{,}000 \textit{ tests}} \times 10,000$
    - $\approx 2.5 \textit{ critical test values per } 10,000 \textit{ tests}$
    - $\frac{3 \textit{ critical test values}}{12{,}000 \textit{ tests}} \times 100,000$
    - $= 25 \textit{ critical test values per } 100,000 \textit{ tests}$

Scenario 11

A district with 2,400 live births in a year recorded 19 infant deaths that occurred before the infants turned 28 days old.

- **Correct measure:**
  - *NEONATAL DEATH RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Deaths\ under\ 28\ days\ old}{Live\ births} \times 1,000$
- **Computation & answer:**
  - $\frac{19\ deaths\ under\ 28\ days\ old}{2{,}400\ live\ births} \times 1,000$
  - $\approx 7.9\ per\ 1,000\ live\ births$

Scenario 12

A town has 8,000 women aged 15 to 49 years and recorded 480 live births in the past year.

- **Correct measure:**
  - *GENERAL FERTILITY RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Live\ births}{Women\ aged\ 15-49} \times 1,000$
- **Computation & answer:**
  - $\frac{480\ live\ births}{8{,}000\ women\ aged\ 15-49} \times 1,000$
  - $= 60\ births\ per\ 1,000\ women\ aged\ 15\ to\ 49$

Scenario 13

A city of 300,000 people recorded 450 deaths due to respiratory disease over one year.

- **Correct measure:**
  - *CAUSE-SPECIFIC DEATH RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Deaths\ from\ one\ named\ cause}{Mid-year\ population} \times 100,000$
- **Computation & answer:**
  - $\frac{450\ deaths\ due\ to\ respiratory\ disease}{300{,}000\ people} \times 100,000$
  - $= 150\ per\ 100,000\ population$

Scenario 14

A school health survey conducted on March 1 finds that 45 of the school's 900 enrolled students currently have scabies.

- **Correct measure:**
  - *POINT PREVALENCE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Existing\ cases\ at\ that\ moment}{Population\ at\ that\ same\ moment} \times 1,000$
- **Computation & answer:**
  - $\frac{45\ scabies\ cases}{900\ students} \times 1,000 = 50\ per\ 1,000\ population$

Scenario 15

During a leptospirosis surveillance period, 340 cases were reported and 12 deaths among leptospirosis patients were logged in that same period.

- **Correct measure:**
  - *DEATH TO CASE RATIO*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Deaths\ from\ disease\ (period)}{New\ cases\ reported\ (same\ period)} \times 100$
- **Computation & answer:**
  - $\frac{12\ deaths\ due\ to\ leptospirosis}{340\ leptospirosis\ cases} \times 100$
  - $\approx 3.5\ deaths\ per\ 100\ cases$

Scenario 16

A municipality with 85,000 residents recorded 680 deaths, from all causes, over the course of one year.

- **Correct measure:**
  - *MORTALITY RATE/CRUDE DEATH RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Total\ deaths}{Mid-year\ population} \times 1,000$
- **Computation & answer:**
  - $\frac{380\ total\ deaths}{85{,}000\ population} \times 1,000$
  - $\approx 4.5\ deaths\ per\ 1,000\ population$

Scenario 17

A 3-year diabetes prevention cohort enrolled 400 adults. Because participants joined and were diagnosed at different times, the study accumulated 1,050 person-years of observation; 63 new diabetes cases were identified.

- **Correct measure:**
  - *INCIDENCE RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{New\ cases\ during\ follow-up}{\Sigma\ person-time\ at\ risk} \times 1,000$
- **Computation & answer:**
  - $\frac{63\ new\ diabetes\ cases}{1{,}050\ person-years} \times 1,000$
  - $= 60\ cases\ per\ 1,000\ person\ years$

Scenario 18

Within a population's 40,000 children aged 0–4 years, 28 deaths (from any cause) occurred over the year.

- **Correct measure:**
  - *AGE-SPECIFIC DEATH RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Deaths\ within\ one\ age\ group}{\Sigma\ person-time\ at\ risk} \times 1,000$
- **Computation & answer:**
  - $\frac{63\ new\ diabetes\ cases}{1{,}050\ \ person-years} \times 1,000$
  - $= 60\ cases\ per\ 1,000\ person\ years$

Scenario 19

Of the same district's 2,400 live births, 11 infants died sometime between 28 days and their first birthday.

- **Correct measure:**
  - *POSTNEONATAL MORTALITY RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Deaths,\ 28\ days\ to\ 1\ year}{Live\ births} \times 1,000$
- **Computation & answer:**
  - $\frac{11\ infant\ deaths}{2{,}400\ live\ births} \times 1,000$
  - $\approx 4.6\ deaths\ per\ 1,000\ live\ births$

Scenario 20

Of 2,100 live births recorded in a city in one year, 189 weighed less than 2,500 grams at birth.

- **Correct measure:**
  - *LOW-BIRTH-WEIGHT RATIO*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Live\ births\ <2{,}500\ g}{Total\ live\ births} \times 100$
- **Computation & answer:**
  - $\frac{189\ live\ births\ <2{,}500\ g}{2{,}100\ total\ live\ births} \times 100 = 9\%\ low\ birth\ weight$

Scenario 21

A province with a mid-year population of 620,000 recorded 9,300 live births over the year.

- **Correct measure:**
  - *CRUDE BIRTH RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Live\ births}{Mid-year\ population} \times 1,000$
- **Computation & answer:**
  - $\frac{9{,}300\ live\ births}{620{,}000\ mid-year\ population} \times 1,000$
  - $= 15\ births\ per\ 1,000\ population$

Scenario 22

Among 5,500 women aged 25 to 29 years, 660 live births were attributed to mothers in that age band during the year.

- **Correct measure:**
  - *AGE-SPECIFIC FERTILITY RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Births\ to\ women\ in\ one\ age\ group}{Women\ in\ that\ same\ age\ group} \times 1,000$
- **Computation & answer:**
  - $\frac{660\ births\ to\ women\ aged\ 25\ to\ 29\ yrs}{5{,}500\ women\ aged\ 25\ to\ 29\ yrs} \times 1,000$
  - $= 120\ births\ per\ 1,000\ women\ aged\ 25\ to\ 29$

Scenario 23

Of 5,000 live births in a region, 40 infants died before reaching their first birthday (28 during the neonatal period, 12 afterward).

- **Correct measure:**
  - *INFANT MORTALITY RATE*
- **Formula (numerator / denominator × multiplier):**
  - $\frac{Deaths\ under\ one\ year\ old}{Live\ births} \times 1,000$
- **Computation & answer:**
  - $\frac{28\ neonatal\ deaths + 12\ postneonatal\ deaths}{5{,}000\ live\ births} \times 1,000$
  - $= 8\ per\ 1,000\ live\ births$

EXTRA NOTE!
According to sir Miasco, it's not necessary to memorize the multipliers (x), because he will not put choices that are only different in multipliers and what matters more are the right numerator and denominator.